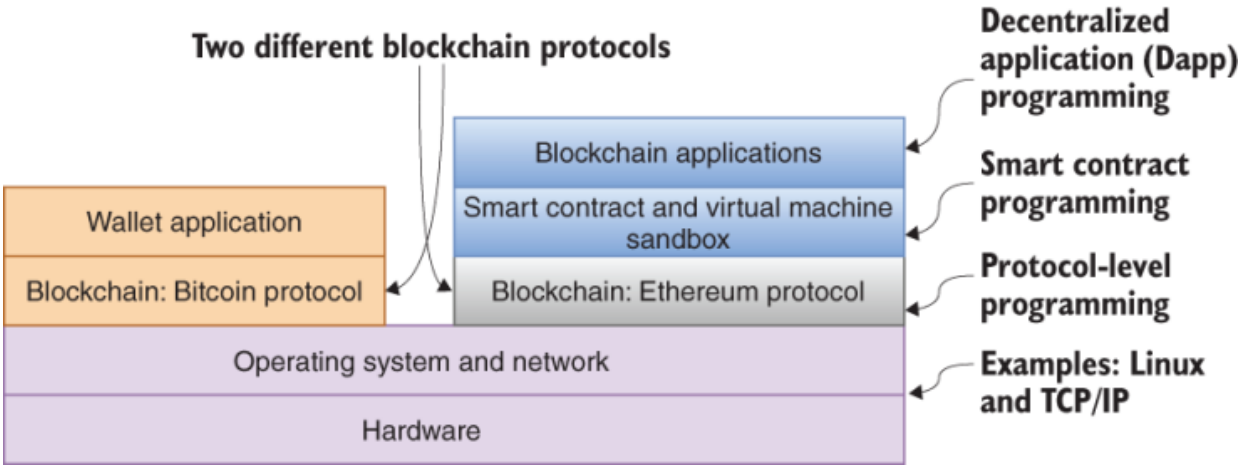


Figure 1.4 Blockchain stacks and types of programming

Figure 1.4 also shows the three levels of programming:

- *Protocol-level programming* --This level involves software that is needed for the deployment and operation of the blockchain itself. This software is similar to your operating system or networking software. If you are a systems programmer and administrator, you'll program at this level. This text does not cover protocol-level programming.
- *Smart contract-level programming* --One level above is smart contract (or rules engine) programming. It is at this level that you design and program the rules for verification and validation, and specify the data and messages that are to be recorded on the underlying blockchain. The smart contract is the engine that drives the blockchain on behalf of the user application. In chapters 2-4, you'll learn in depth about the design, development, and testing of smart contracts.